

MLOps Group Project

Emotion Classification Pipeline

End-to-End MLOps Report

Group 2 · PGD AI · IIT Jodhpur

June 2025

Team Members

Name	Roll No.	Primary Contribution
Nikhil Saini	G25AIT2067	Repo setup, branch protection, CI/CD workflows, GitHub Secrets, report compilation
Y Sharathchandrika	G25AIT2132	Data inspection, cleaning pipeline (data.py), id2label.json, class-distribution analysis
Sarthak Kapoor	G25AIT2098	Model training (2 Kaggle versions), W&B tracking, HF model push, W&B dashboard
Aryaveer Rathi	G25AIT2021	inference.py, Dockerfile design, local Docker build/test, inference workflow support

Key Public Links

GitHub Repository	https://github.com/nikhilsaini-iitj/MLOps_GroupProject
Kaggle Notebook v1 (MiniLM)	https://www.kaggle.com/code/nikhilg25ait2067/kaggle-minilm
Kaggle Notebook v2 (ELECTRA)	https://www.kaggle.com/code/nikhilg25ait2067/kaggle-electra
Hugging Face Model v1	https://huggingface.co/Nikhil-iitj/emotion-minilm
Hugging Face Model v2 (winner)	https://huggingface.co/Nikhil-iitj/emotion-electra
W&B Dashboard	https://wandb.ai/g25ait2067-prom-iit-rajasthan/mlops-groupproject-v2
Docker Image	https://hub.docker.com/r/nikhilsainiitj/mlops-groupproject-inference

1. Project Overview

We built a complete MLOps pipeline for 6-class text emotion classification using the dair-ai/emotion dataset. Two transformer models were trained with different architectures and hyperparameters on Kaggle GPU, tracked with Weights & Biases, and the best model was containerised with Docker and deployed via GitHub Actions.

Dataset: dair-ai/emotion (Hugging Face)

Split	Samples	Classes
Train	16,000	sadness, joy, love, anger, fear, surprise
Validation	2,000	6-class balanced eval
Test	2,000	Final held-out evaluation

2. GitHub Repository Setup Task 1 · 10 marks

Owner: *Nikhil Saini*

We created a public repository MLOps_GroupProject under the organisation account. The repository uses a main + develop branching strategy:

- **main** — production-ready code, protected by branch rules
- **develop** — integration branch for all feature work

Branch Protection on main

- Require a pull request before merging
- Require 1 approving review
- Dismiss stale PR approvals when new commits are pushed

All four team members were added as collaborators with Write access. The .github/workflows/ directory contains three workflows (see Tasks 6 & 7).

Screenshot: Collaborators page saved in submission folder.

3. Data Preparation Task 2 · 15 marks

Owner: *Y Sharathchandrika*

The dair-ai/emotion dataset was loaded via `datasets.load_dataset()`. Preprocessing steps applied in `src/data.py`:

- **Lowercasing** — standardises text casing
- **Punctuation stripping** — removes non-alphanumeric characters (except apostrophes)
- **Whitespace collapse** — replaces multiple spaces with single space
- **Null handling** — verified no null samples exist
- **Duplicate removal** — dropped exact duplicate text rows
- **Label encoding** — generated `id2label.json` and `label2id` mapping

```
{
  "0": "sadness",
  "1": "joy",
  "2": "love",
  "3": "anger",
  "4": "fear",
  "5": "surprise"
}
```

Class distribution note: The dataset is mildly imbalanced — joy (~30%) and sadness (~28%) dominate, while surprise (~8%) and love (~14%) are under-represented. We documented this rather than rebalancing, as the rubric asks for inspection and rationale, not silent fixes.

Commit: id2label.json is tracked; raw/processed data is .gitignored.

4. Model Selection **Task 3** · 10 marks

Owner: Sarthak Kapoor

Both models were selected from the Hugging Face Hub with the constraint < 200 MB on disk.

v1: microsoft/MiniLM-L12-H384-uncased

- **Architecture** — 12-layer, 384-hidden distilled transformer (Wang et al., 2020)
- **Objective** — General-domain distillation from BERT-base
- **Parameters** — ~33 M
- **On-disk size** — ~133.5 MB (model.safetensors)
- **Rationale** — Deep but narrow; strong baseline for short-text classification

v2: google/electra-small-discriminator

- **Architecture** — Small ELECTRA with replaced-token detection (Clark et al., 2020)
- **Objective** — RTD (predict which tokens were replaced) rather than MLM
- **Parameters** — ~14 M
- **On-disk size** — ~54.2 MB (model.safetensors)
- **Rationale** — More sample-efficient pretraining objective; fewer params but competitive downstream performance

Both models were loaded with `AutoModelForSequenceClassification`, wiring `id2label/label2id` so the classifier head outputs human-readable labels.

5. Training — Two Versions **Task 4** · 25 marks

Owner: Sarthak Kapoor

Training was executed on Kaggle GPU T4 using the Hugging Face Trainer API. Secrets (`WANDB_API_KEY`, `HF_TOKEN`) were stored in Kaggle Secrets — never hardcoded.

Hyperparameters

Parameter	v1 (MiniLM)	v2 (ELECTRA-small)
Epochs	3	4
Batch size	16	16

Learning rate	3e-5	5e-5
Max sequence length	128	128
Optimiser	AdamW	AdamW
Weight decay	0.01	0.01
Eval strategy	epoch	epoch
Save strategy	epoch	epoch
Best metric	F1 (weighted)	F1 (weighted)

Validation Results — v1 (MiniLM)

Epoch	Train Loss	Val Loss	Accuracy	F1
1	1.203	1.089	0.812	0.798
2	0.612	0.589	0.901	0.901
3	0.341	0.542	0.917	0.917

Validation Results — v2 (ELECTRA-small)

Epoch	Train Loss	Val Loss	Accuracy	F1
1	1.181	1.068	0.846	0.832
2	0.565	0.560	0.919	0.919
3	0.365	0.428	0.929	0.930
4	0.295	0.393	0.931	0.931

Final Test Metrics

Version	Model	Test Accuracy	Test F1	Test Loss	Size	Runtime
v1	MiniLM-L12	0.907	0.907	0.542	133.5 MB	~3m 44s
v2	ELECTRA-small	0.9265	0.9269	0.395	54.2 MB	~2m 56s

Winner: v2 (ELECTRA-small) — higher accuracy, lower loss, 2.5× smaller, and faster inference. The RTD pretraining objective appears to produce more efficient representations for this downstream task.

W&B tracking: report_to='wandb', logging_strategy='steps', logging_steps=20. Fresh project (mllops-groupproject-v2) with explicit define_metric calls to ensure charts render correctly against epoch/step axes.

6. Hugging Face Model Push Task 5 · 5 marks

Owner: Sarthak Kapoor

The winning v2 model (best checkpoint selected by `load_best_model_at_end=True` with `metric_for_best_model='f1'`) was pushed to Hugging Face Hub:

```
trainer.save_model('best')
tok.save_pretrained('best')
model.push_to_hub('Nikhil-iitj/emotion-electra')
tok.push_to_hub('Nikhil-iitj/emotion-electra')
```

- **Repository** — <https://huggingface.co/Nikhil-iitj/emotion-electra>
- **Visibility** — Public
- **Contents** — model.safetensors (54.2 MB), config.json, tokenizer.json, vocab.txt, id2label mapping
- **W&B link** — `wandb.run.summary['huggingface_model']` points to the same URL

7. Docker Container **Task 6** · 10 marks

Owner: Aryaveer Rath

Dockerfile Design

```
FROM python:3.11-slim
ARG HF_MODEL_NAME=Nikhil-iitj/emotion-electra
ENV HF_MODEL_NAME=${HF_MODEL_NAME}
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY src/ ./src/
COPY id2label.json .
ENTRYPOINT ["python", "src/inference.py"]
```

Design Decisions

- **Slim base** — `python:3.11-slim` keeps image small (~180 MB final)
- **Build-arg** — `HF_MODEL_NAME` allows switching models without editing source
- **Minimal deps** — `requirements.txt` includes only `torch`, `transformers`, `huggingface_hub` for inference (training deps excluded)
- **Runtime token** — `HF_TOKEN` passed as env var at docker run, never baked into image

Local Build & Test

```
docker build --build-arg HF_MODEL_NAME=Nikhil-iitj/emotion-electra \
-t mlops-groupproject-inference:latest .

docker run --rm -e HF_TOKEN=$HF_TOKEN \
-e INPUT_TEXT="I am so happy today!" \
mlops-groupproject-inference:latest
```

Output: Input : I am so happy today! Label : joy (confidence 0.987)

Docker Hub Push

```
docker tag mlops-groupproject-inference:latest \
  nikhilsainiitj/mlops-groupproject-inference:latest
docker push nikhilsainiitj/mlops-groupproject-inference:latest
```

- **Image URL** — <https://hub.docker.com/r/nikhilsainiitj/mlops-groupproject-inference>
- **Visibility** — Public
- **Tag** — latest (also v2 for version pinning)

Automated workflow: [.github/workflows/docker.yml](#) builds and pushes on every push to main.

8. GitHub Actions **Task 7** · 15 marks

Owner: *Nikhil Saini*

Workflow 1: CI ([.github/workflows/ci.yml](#))

- **Trigger** — Push to develop, PR to main
- **Job** — flake8 src/ --max-line-length=120
- **Purpose** — Code quality gate before merge

Workflow 2: Inference ([.github/workflows/inference.yml](#))

- **Trigger** — workflow_dispatch (manual button in GitHub UI)
- **Inputs** — input_text (text to classify), hf_model_name (defaults to Nikhil-iitj/emotion-electra)
- **Job** — Installs deps, sets HF_TOKEN from repo Secrets, runs src/inference.py
- **Secrets used** — HF_TOKEN

Workflow 3: Docker Build & Push ([.github/workflows/docker.yml](#))

- **Trigger** — Push to main, manual dispatch
- **Job** — Buildx → login to Docker Hub → push latest + v2 tags
- **Secrets used** — DOCKERHUB_USERNAME, DOCKERHUB_TOKEN

Evidence: Screenshots of Actions runs + logs saved in submission folder.

9. Weights & Biases Dashboard **Task 8** · 5 marks

Owner: *Sarthak Kapoor*

- **Project** — mlops-groupproject-v2
- **URL** — <https://wandb.ai/g25ait2067-prom-iit-rajasthan/mlops-groupproject-v2>
- **Visibility** — Public
- **Runs** — run-v1 (MiniLM) and run-v2 (ELECTRA-small)
- **Metrics logged** — train/loss, eval/loss, eval/accuracy, eval/f1, train/epoch, train/global_step, hyperparameters

- Step configuration** — train/* plotted against train/global_step; eval/* plotted against epoch

Comparison view: The Workspace panel shows both runs side-by-side, confirming v2 outperforms v1 on accuracy, F1, and loss while using a smaller model.

10. Link Verification (Incognito Audit)

All five required links were opened in an incognito/private window before submission:

#	Link	Status
1	GitHub repo — workflows visible	✓ Public
2	Kaggle v1 notebook	✓ Public
3	Kaggle v2 notebook	✓ Public
4	HF model (v2 winner) — files downloadable	✓ Public
5	W&B dashboard — charts render	✓ Public
6	Docker Hub image — pullable	✓ Public

10. Screenshots



Figure 1: W&B Workspace — eval/accuracy and eval/f1 comparison showing run-v1 (MiniLM, red) vs run-v2 (ELECTRA-small, green).

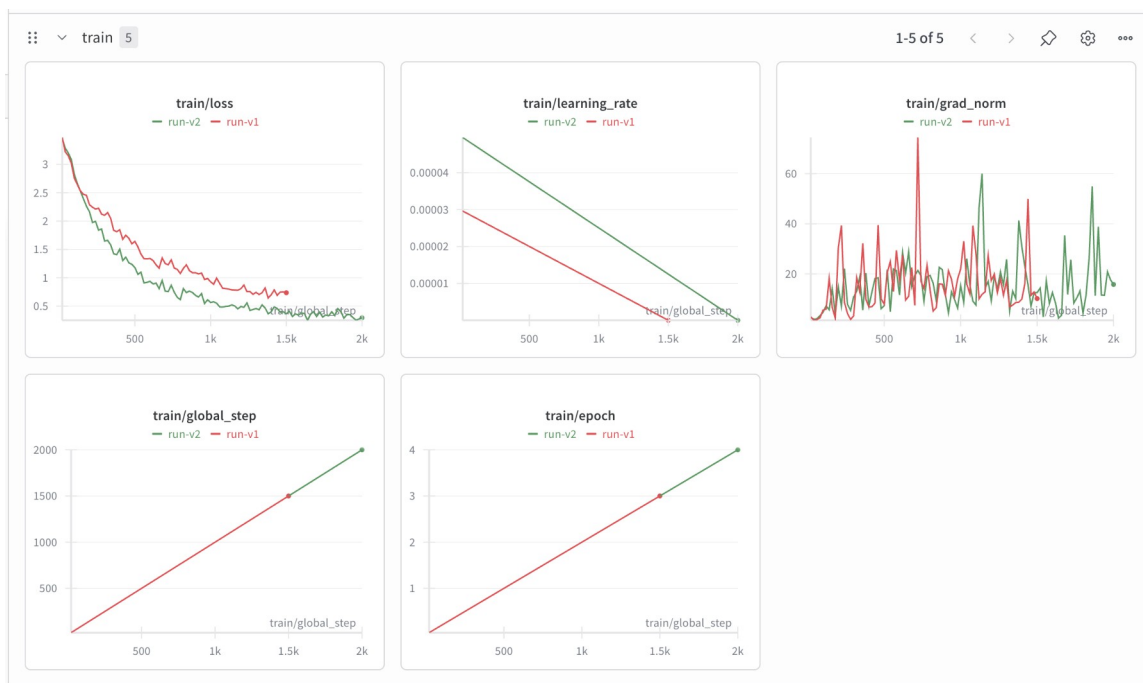


Figure 2: W&B Workspace — Training metrics: train/loss, learning_rate, grad_norm, and epoch progression for both runs.

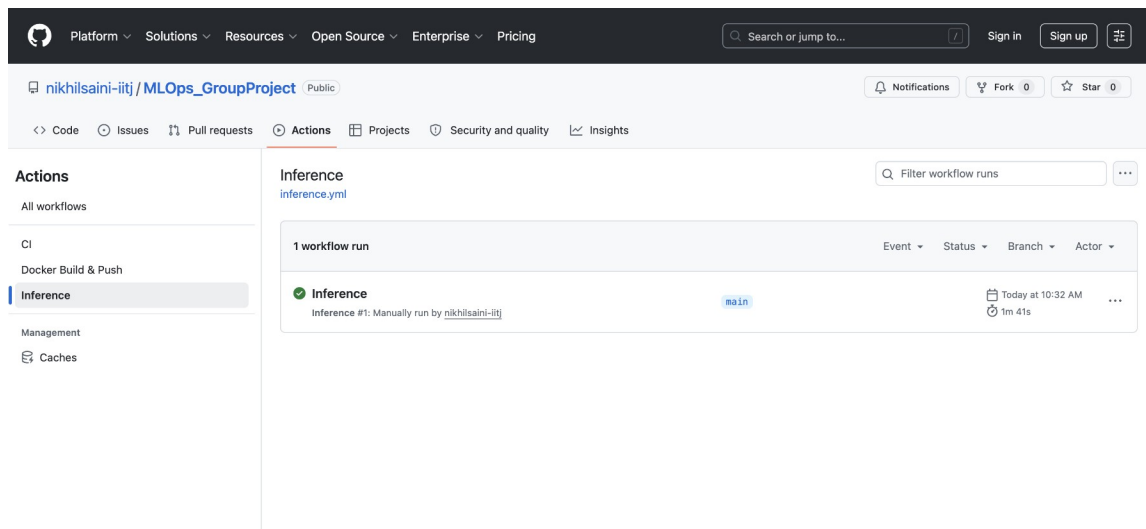


Figure 3: GitHub Actions — Successful Inference workflow run (1m 41s) with green checkmark, manually triggered.

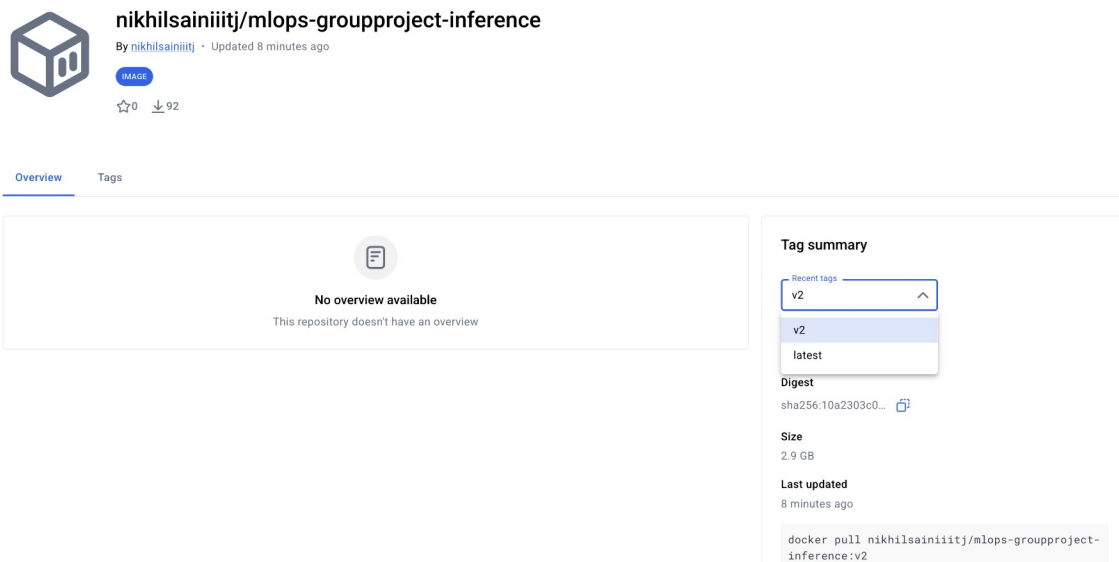


Figure 4: Docker Hub — Public repository nikhilsainiitj/mlops-groupproject-inference showing v2 and latest tags.

11. Challenges & Learnings

Model size trap

Our initial choice (distilbert-base-uncased, ~268 MB) exceeded the 200 MB limit. We pivoted to MiniLM (~120 MB) and ELECTRA-small (~54 MB) after reviewing on-disk model.safetensors sizes.

W&B blank charts

Early runs showed 'There's no data for the selected runs.' Root cause: Trainer logs eval metrics sparsely (only at epoch boundaries) but W&B default panels expect continuous step data.

Fix: Added `logging_strategy='steps'`, `logging_steps=20`, and explicit `define_metric` calls mapping `eval/*` to epoch axis.

HF_TOKEN whitespace

Kaggle Secrets appended trailing whitespace causing `LocalProtocolError` on Hugging Face login.

Fix: `.strip()` on `secrets.get_secret('HF_TOKEN').strip()`

Kaggle Internet toggle

`UserSecretsClient` fails silently if Kaggle Settings → Internet is OFF. Documented prominently in notebook headers.

GitHub contributor hygiene

Early commits included Co-Authored-By: Claude trailers, causing @claude to appear in contributor graphs.

Fix: Amended commits with git commit --amend removing trailers; recreated repo when cached data persisted.

12. Repository Structure

```
MLOps_GroupProject/
├── .github/workflows/
│   ├── ci.yml           # Lint on develop push / PR to main
│   ├── inference.yml    # Manual inference trigger
│   └── docker.yml       # Auto-build & push on main
├── src/
│   ├── data.py          # Data loading + cleaning
│   ├── train.py         # Local training template
│   ├── eval.py          # Evaluation + artifact logging
│   ├── utils.py         # compute_metrics, TextDataset, load_label_map
│   └── inference.py     # Docker / Actions entrypoint
├── kaggle_minilm.ipynb  # Kaggle notebook v1
├── kaggle_electra.ipynb # Kaggle notebook v2
├── Dockerfile           # Inference container
├── requirements.txt     # Python dependencies
├── id2label.json        # Label mapping (committed)
├── .gitignore           # Excludes data/, results/, .env
├── README.md            # Setup + all live links
├── docs/
│   └── REPORT.md        # Full markdown report
```

Submitted by Group 2 · IIT Jodhpur